

Open Questions on the Bernoulli Factory Problem

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1 Background

Suppose there is a coin that shows heads with an unknown probability, λ . The goal is to use that coin (and possibly also a fair coin) to build a “new” coin that shows heads with a probability that depends on λ , call it $f(\lambda)$. This is the *Bernoulli factory problem*, and it can be solved for a function $f(\lambda)$ only if it’s continuous. (For example, flipping the coin twice and taking heads only if exactly one coin shows heads, the probability $2\lambda(1 - \lambda)$ can be simulated.)

This page contains several questions about the **Bernoulli factory**¹ problem. Answers to them will greatly improve my pages on this site about Bernoulli factories. If you can answer any of them, post an issue in the **GitHub issues page**².

Note: The Bernoulli factory problem is a special case of a more general mathematical problem that I call “**The Sampling Problem**”³.

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3 Polynomials that approach a Bernoulli factory function “fast”

This question involves solving the Bernoulli factory problem with polynomials.⁴

¹<https://peteroupc.github.io/bernoulli.html>

²<https://github.com/peteroupc/peteroupc.github.io/issues>

³<https://peteroupc.github.io/sampling.html>

⁴See also the following questions on *Mathematics Stack Exchange* and *MathOverflow*: **Converging polynomials, Error bounds, A conjecture, Lorentz operators, Series representations**. <https://math.stackexchange.com/questions/3904732/what-are-ways-to-compute-polynomials-that-converge-from-above-and-below-to-a-con> <https://mathoverflow.net/questions/442057/explicit-and-fast-error-bounds-for-approximating-continuous-functions> <https://mathoverflow.net/questions/427595/a-conjecture-on-consistent-monotone-sequences-of-polynomials-in-bernstein-form> <https://mathoverflow.net/questions/427595/a-conjecture-on-consistent-monotone-sequences-of-polynomials-in-bernstein-form>

In this question, a polynomial $P(x)$ is written in *Bernstein form of degree n* if it is written as—

$$P(x) = \sum_{k=0}^n a_k \binom{n}{k} x^k (1-x)^{n-k},$$

where the real numbers $a_0, \dots, a_n$ are the polynomial's *Bernstein coefficients*.

The degree- n *Bernstein polynomial* of an arbitrary function $f(x)$ has Bernstein coefficients $a_k = f(k/n)$. In general, this Bernstein polynomial differs from f even if f is a polynomial.

3.1 Main Question

Suppose $f : [0, 1] \rightarrow [0, 1]$ is continuous and belongs to a large class of functions (for example, the k -th derivative, $k \geq 0$, is continuous, Lipschitz continuous, concave, strictly increasing, or bounded variation, or f is real analytic).

1. (*Exact Bernoulli factory*): Compute the Bernstein coefficients of a sequence of polynomials (g_n) of degree $2, 4, 8, \dots, 2^i, \dots$ that converge to f from below and satisfy: $(g_{2n} - g_n)$ is a polynomial with nonnegative Bernstein coefficients once it's rewritten to a polynomial in Bernstein form of degree exactly $2n$. Assume $0 < f(\lambda) < 1$ or a piecewise polynomial can come between f or $1 - f$ and the x -axis.
2. (*Approximate Bernoulli factory*): Given $\epsilon > 0$, compute the Bernstein coefficients of a polynomial or rational function (of some degree n) that is within ϵ of f .⁵
3. (*Series expansion of simple functions*): Find a nonnegative random variable X and a series $f(\lambda) = \sum_{a \geq 0} \gamma_a(\lambda)$ such that $\gamma_a(\lambda)/\mathbb{P}(X = a)$ (letting $0/0$ equal 0) is a polynomial or rational function with rational Bernstein coefficients lying in $[0, 1]$.⁶ Do the same for a function that is within ϵ of f , rather than f .

The convergence rate must be $O(1/n^{r/2})$ if the class has only functions with Lipschitz-continuous $(r-1)$ -th derivative. (Emphasis is given to cases where $r \geq 3$.) The method may not introduce transcendental or trigonometric functions (as with Chebyshev interpolants).

3.2 Solving the Bernoulli factory problem with polynomials

An **algorithm**⁷ (Łatuszyński et al. 2009/2011)⁸ simulates a function that admits a Bernoulli factory $f(\lambda)$ via two sequences of polynomials that converge from above and below to that function. Roughly speaking, the algorithm works as follows:

1. Generate U , a uniform random variate in $[0, 1]$.
2. Flip the input coin (with a probability of heads of λ), then build an upper and lower bound for $f(\lambda)$, based on the outcomes of the flips so far. In this case, these bounds come from two degree- n polynomials that approach f as n gets large, where n is the number of coin flips so far in the algorithm.

<https://mathoverflow.net/questions/407179/using-the-holtz-method-to-build-polynomials-that-converge-to-a-continuous-function>
<https://mathoverflow.net/questions/409174/concave-functions-series-representation-and-converging-polynomials>

⁵As discussed later, the Bernstein polynomials solve this question for functions no “smoother” than arbitrary functions in $C^2[0, 1]$, but their convergence rate is $O(1/n)$ in general. For some of these functions, there may be rational-function approximations or more practical methods with a faster convergence rate. By contrast, results for functions “smoother” than C^2 are hard to find.

⁶An example of X is $\mathbb{P}(X = a) = p(1-p)^a$ where $0 < p < 1$ is a known rational. This question's requirements imply that $\sum_{a \geq 0} \max_{\lambda} \text{abs}(\gamma_a(\lambda)) \leq 1$. The proof of Keane and O'Brien (1994) produces a convex combination of polynomials with 0 and 1 as Bernstein coefficients, but the combination is difficult to construct (it requires finding maximums, for example) and so this proof does not appropriately answer this question.

⁷https://peteroupc.github.io/bernoulli.html#General_Factory_Functions

⁸Łatuszyński, K., Kosmidis, I., Papaspiliopoulos, O., Roberts, G.O., “**Simulating events of unknown probabilities via reverse time martingales**”, arXiv:0907.4018v2 [stat.CO], 2009/2011; Random Struct. Alg., 38: 441-452 (2011). <https://doi.org/10.1002/rsa.20333> <https://arxiv.org/abs/0907.4018v2>

3. If U is less than or equal to the lower bound, return 1. If U is greater than the upper bound, return 0. Otherwise, go to step 2.

The result of the algorithm is 1 with probability *exactly* equal to $f(\lambda)$, or 0 otherwise.

However, the algorithm requires the polynomial sequences to meet certain requirements, one of which is:

For $f(\lambda)$ there must be a sequence of polynomials (g_n) in Bernstein form of degree 1, 2, 3, ... that converge to f from below and satisfy: $(g_{n+1} - g_n)$ is a polynomial with nonnegative Bernstein coefficients once it's rewritten to a polynomial in Bernstein form of degree exactly $n + 1$ (Nacu and Peres (2005)⁹; Holtz et al. (2011)¹⁰).¹¹ For $f(\lambda) = 1 - f(\lambda)$ there must likewise be a sequence of this kind.

3.3 A Matter of Efficiency

However, ordinary Bernstein polynomials converge to a function at the rate $\Omega(1/n)$ unless the function is linear, a result known since Voronovskaya (1932)¹² and a rate that will lead to an **infinite expected number of coin flips in general**. (See also my **supplemental notes**¹³.)

But Lorentz (1963)¹⁴ showed that if the function is positive and has a continuous k -th derivative, there are polynomials with nonnegative Bernstein coefficients that converge at the rate $O(1/n^{k/2})$ (and thus can enable a **finite expected number of coin flips** if the function is “smooth” enough; for example, if the function’s second or higher-order derivative is Lipschitz continuous).

Thus, researchers have studied alternatives to Bernstein polynomials that improve the convergence rate for “smoother” functions. See Holtz et al. (2011)¹⁵, Sevy (1991)¹⁶, Waldron (2009)¹⁷, Costabile et al. (2005)¹⁸, Han (2003)¹⁹, Khosravian-Arab et al. (2018)²⁰, and references therein; see also Micchelli (1973)²¹, Güntürk

⁹Nacu, Șerban, and Yuval Peres. “Fast simulation of new coins from old”, The Annals of Applied Probability 15, no. 1A (2005): 93-115. <https://doi.org/10.1214/105051604000000549>

¹⁰Holtz, O., Nazarov, F. & Peres, Y. New Coins from Old, Smoothly. Constr Approx 33, 331–363 (2011). <https://doi.org/10.1007/s00365-010-9108-5>

¹¹The condition on nonnegative Bernstein coefficients ensures that not only the polynomials “increase” to $f(\lambda)$, but also their Bernstein coefficients. This condition is equivalent in practice to the following statement (Nacu & Peres 2005). For every integer $n \geq 1$ that’s a power of 2, $a(2n, k) \geq \mathbb{E}[a(n, X_{n,k})] = \left(\sum_{i=0}^k a(n, i) \binom{n}{i} \binom{n}{k-i} / \binom{2n}{k} \right)$, where $a(n, k)$ is the degree- n polynomial’s k -th Bernstein coefficient, where $0 \leq k \leq 2n$ is an integer, and where $X_{n,k}$ is a hypergeometric($2n, k, n$) random variable. A hypergeometric($2n, k, n$) random variable is the number of “good” balls out of k balls taken uniformly at random, all at once, from a bag containing $2n$ balls, n of which are “good”.

¹²E. Voronovskaya, “Détermination de la forme asymptotique d’approximation des fonctions par les polynômes de M. Bernstein”, 1932.

¹³<https://peteroupc.github.io/bernsupp.html>

¹⁴Lorentz, G.G. The degree of approximation by polynomials with positive coefficients. Math. Ann. 151, 239–251 (1963). <https://doi.org/10.1007/BF01398235>

¹⁵Holtz, O., Nazarov, F. & Peres, Y. New Coins from Old, Smoothly. Constr Approx 33, 331–363 (2011). <https://doi.org/10.1007/s00365-010-9108-5>

¹⁶Sevy, J., “Acceleration of convergence of sequences of simultaneous approximants”, dissertation, Drexel University, 1991. <https://doi.org/10.17918/00010296>

¹⁷Waldron, S., “Increasing the polynomial reproduction of a quasi-interpolation operator”, Journal of Approximation Theory 161 (2009). <https://doi.org/10.1016/j.jat.2008.08.011> <https://www.sciencedirect.com/science/article/pii/S0021904508001640>

¹⁸Costabile, F., Gualtieri, M.I., Serra, S., “Asymptotic expansion and extrapolation for Bernstein polynomials with applications”, BIT 36 (1996). <https://doi.org/10.1007/BF01733787>

¹⁹Han, Xuli. “Multi-node higher order expansions of a function.” Journal of Approximation Theory 124.2 (2003): 242-253. <https://doi.org/10.1016/j.jat.2003.08.001>

²⁰Khosravian-Arab, H., Dehghan, M. & Eslahchi, M.R. A new approach to improve the order of approximation of the Bernstein operators: theory and applications. Numerical Algorithms 77, 111–150 (2018). <https://doi.org/10.1007/s11075-017-0307-z>.

²¹Micchelli, Charles. “The saturation class and iterates of the Bernstein polynomials”, Journal of Approximation Theory 8, no. 1 (1973): 1-18. <https://doi.org/10.1016/0021-9045%2873%2990028-2> <https://www.sciencedirect.com/science/article/pii/0021904573900282>

and Li (2021a)²², (2021b)²³, Draganov (2024)²⁴, and Tachev (2022)²⁵.

These alternative polynomials usually come with results where the error bound is the desired $O(1/n^{k/2})$, but most of those results (with the notable exception of Sevy) have hidden constants with no upper bounds given, making them unimplementable (that is, it can't be known beforehand whether a given polynomial will come close to the target function within a user-specified error tolerance).

3.4 A Conjecture on Polynomial Approximation

The following is a **conjecture**²⁶ that could help reduce this problem to the problem of finding explicit error bounds when approximating a function by polynomials.

Let $f(\lambda) : [0, 1] \rightarrow (0, 1)$ have a continuous r -th derivative, where $r \geq 1$, let $M = \max_{0 \leq i \leq r, 0 \leq \lambda \leq 1} |f^{(r)}(\lambda)|$, let B_n be the degree- n Bernstein operator, and let $(W_n(g; \lambda))_n$ be a sequence of operators that map a continuous function g on $[0, 1]$ to a bounded function on $[0, 1]$ and converge uniformly to f .

Prove or disprove the following statement. For each integer $n \geq 1$, suppose that there is $D > 0$ such that $|f(\lambda) - B_n(W_n(f; \lambda); \lambda)| \leq DM/n^{r/2}$ whenever $0 \leq \lambda \leq 1$. Then there is $C_1 > 0$ depending only on r and $(W_n)_n$ such that the inequality $\sum_{i=0}^n (W_{2n}(f; \frac{k}{2n}) - W_n(f; \frac{i}{n})) \sigma_{n,k,i} \geq -C_1 M/n^{r/2}$ holds true for each integer $n \geq 1$ and whenever $0 \leq k \leq 2n$, where $\sigma_{n,k,i} = \binom{n}{i} \binom{n}{k-i} / \binom{2n}{k}$. Call the latter inequality (PB).²⁷

Also, prove or disprove the same statement with the assumption that n is a power of 2. When C_0 exists, find a good upper bound for it.

Notes:

1. I believe there is a counterexample to this conjecture, namely the sequence $B_n(W_n(f; \lambda); \lambda) = \frac{(T_n(1-2\lambda)+1)\varphi_n}{2\mu_n} + 1/2$, where φ_n is a decreasing sequence of positive numbers that tends slowly enough to 0, μ_n is the maximum Bernstein coefficient (in absolute value) of the degree- n polynomial $(T_n(1-2\lambda)+1)/2$, and $T_n(x)$ is the Chebyshev polynomial of the first kind of degree n . W_n then maps to a piecewise linear function that connects the Bernstein coefficients of $B_n(W_n(\lambda))$, so that (W_n) is a sequence of operators that converges at an arbitrarily slow rate (depending on φ_n) to the constant $1/2$. $B_n(W_n(\lambda))$ converges uniformly, at an exponential rate, to $f(\lambda) = 1/2$, so that $M = 1/2$. If this counterexample is valid, the conjecture may still be true with an additional assumption on the convergence rate of W_n , say, $O(1/n)$ or $O(1/n^{r/2})$ or $O(1/n^{(r-1)/2})$.
2. If $W_n(f)$ is a linear operator, the left-hand side of (PB) can be treated as a linear operator, too, after a change of variables from k to $2n\lambda$, with $0 \leq \lambda \leq 1$. Call the new operator $L_n(f)$. The goal is then to find an upper bound for L_n that is $O(1/n^{r/2})$. See the example given in the next section.

²²Güntürk, C. Sinan, and Weilin Li. “Approximation with one-bit polynomials in Bernstein form”, arXiv:2112.09183 (2021); Constr Approx 57, 601–630 (2023). <https://doi.org/10.1007/s00365-022-09608-y> <https://arxiv.org/pdf/2112.09183>

²³Güntürk, C. Sinan, and Weilin Li. “Approximation of functions with one-bit neural networks”, arXiv:2112.09181 (2021). <https://arxiv.org/abs/2112.09181>

²⁴Draganov, B.R., “Simultaneous approximation by the Bernstein operator”, dissertation, Sofia University “St. Kliment Ohridski”, 2024. https://www.fmi.uni-sofia.bg/sites/default/files/dissertation_work_doctor_of_science/dissdsci_borislavdraganov.pdf

²⁵Tachev, Gancho. “Linear combinations of two Bernstein polynomials”, *Mathematical Foundations of Computing*, 2022. <https://doi.org/10.3934/mfc.2022061>

²⁶https://peteroupc.github.io/bernsupp.html#A_Conjecture_on_Polynomial_Approximation

²⁷If the inequality holds, then there is $C_0 > 0$ such that the polynomials (g_n) in Bernstein form of degree 2, 4, 8, ..., 2^i , ..., defined as $g_n = B_n(W_n(f; \lambda); \lambda) - C_0 M/n^{r/2}$, converge from below to f and satisfy: $(g_{2n} - g_n)$ is a polynomial with nonnegative Bernstein coefficients once it's rewritten to a polynomial in Bernstein form of degree exactly $2n$; see Nacu and Peres (2005)[⁵]. $\sigma_{n,k,i}$ is the probability that a hypergeometric($2n, k, n$) random variable equals i .

3. For every $s \geq 2$, if the sequence (W_n) is such that $B_n(W_n(f))$ preserves all polynomials of degree s or less, then $B_n(W_n(f))$ must be nonlinear, nonpositive, or equal to f (Gavrea and Ivan 2018²⁸, Guessab et al. 2009²⁹), so that, for example, (W_n) cannot be a sequence of linear positive operators (Finta 2013)³⁰.
4. For more notes on finding error bounds, see “**Notes on Approximation Theory**”³¹. That page also has an example of finding error bounds in the case $W_n = 2f - B_n(f)$, which corresponds to the iterated Boolean sum of Bernstein polynomials, with one iteration (Güntürk and Li 2021a, Theorem 5)³².

3.5 Strategies

The following are some strategies for answering these questions:

- Verify my proofs for the results on error bounds for certain polynomials in “**Results Used in Approximations By Polynomials**”³³, including:
 - Iterated Boolean sums (linear combinations of iterates) of Bernstein polynomials $(B_n(W_n) = f - (f - B_n(f))^k)$:³⁴ Propositions B10C and B10D.
 - Linear combinations of Bernstein polynomials (see Costabile et al. (2005)³⁵): Proposition B10.
 - The **Lorentz operator**³⁶ (Holtz et al. 2011)³⁷.
- Find the hidden constants θ_α , s , and D as well as those in Lemmas 15, 17 to 22, 24, and 25 in Holtz et al. (2011)³⁸.
- Find operators or functions of the following kinds and find explicit bounds, with no hidden constants, on the approximation error:
 - Operators that produce a degree- n polynomial in Bernstein form, or a ratio of two such polynomials, such that—
 - * the operator preserves polynomials at a higher degree than linear functions, or
 - * $O(n^2)$ sample points are required.
 - Operators that produce polynomials from samples at *rational* values of a function f that cluster at a quadratic rate toward the endpoints (Adcock et al. 2019)³⁹ (for example, values that converge to Chebyshev points $\cos(j\pi/n)$ with increasing n , or to Legendre points). See also 7, 8, and 12 of

²⁸Ioan Gavrea, Mircea Ivan, “A note on the fixed points of positive linear operators”, Journal of Approximation Theory (227), 2018, <https://doi.org/10.1016/j.jat.2017.12.001>.

²⁹Guessab, A., Nouisser, O. & Schmeisser, G. Enhancement of the algebraic precision of a linear operator and consequences under positivity. Positivity 13, 693–707 (2009). <https://doi.org/10.1007/s11117-008-2253-4>

³⁰Finta Z., “Bernstein type operators having 1 and x^j as fixed points”, Cent. Eur. J. Math., 11 (2013), pp. 2257–2261. <https://doi.org/10.2478/s11533-013-0310-0>

³¹<https://peteroupc.github.io/approxtheory.html>

³²Güntürk, C. Sinan, and Weilin Li. “**Approximation with one-bit polynomials in Bernstein form**”, arXiv:2112.09183 (2021); Constr Approx 57, 601–630 (2023). <https://doi.org/10.1007/s00365-022-09608-y> <https://arxiv.org/pdf/2112.09183>

³³https://peteroupc.github.io/bernapprox.html#Results_Used_in_Approximations_by_Polynomials

³⁴If $W_n(f; 0) = f(0)$ and $W_n(f; 1) = f(1)$ for every n , then the inequality is automatically true when $k = 0$ and $k = 2n$, so that the statement has to be checked only for $0 < k < 2n$. If, in addition, W_n is symmetric about $1/2$, so that $W_n(f; \lambda) = W_n(f; 1-\lambda)$ whenever $0 \leq \lambda \leq 1$, then the statement has to be checked only for $0 < k \leq n$ (since the values $\sigma_{n,k,i} = \binom{n}{i} \binom{n}{k-i} / \binom{2n}{k}$ are symmetric in that they satisfy $\sigma_{n,k,i} = \sigma_{n,k,k-i}$). Special cases for this question are if $W_n = 2f - B_n(f)$ and r is 3 or 4, or $W_n = B_n(B_n(f)) + 3(f - B_n(f))$ and r is 5 or 6; these cases correspond to the iterated Boolean sum of Bernstein polynomials: $B_n(W_n) = f - (f - B_n(f))^k$ (where the k indicates k -fold nesting).

³⁵Costabile, F., Gualtieri, M.I., Serra, S., “Asymptotic expansion and extrapolation for Bernstein polynomials with applications”, BIT 36 (1996). <https://doi.org/10.1007/BF01733787>

³⁶<https://doi.org/10.1007/s00365-010-9108-5>

³⁷Holtz, O., Nazarov, F. & Peres, Y. New Coins from Old, Smoothly. Constr Approx 33, 331–363 (2011). <https://doi.org/10.1007/s00365-010-9108-5>

³⁸Holtz, O., Nazarov, F. & Peres, Y. New Coins from Old, Smoothly. Constr Approx 33, 331–363 (2011). <https://doi.org/10.1007/s00365-010-9108-5>

³⁹Adcock, B., Platte, R.B., Shadrin, A., “Optimal sampling rates for approximating analytic functions from pointwise samples”, IMA Journal of Numerical Analysis 39(3), July 2019. <https://doi.org/10.1093/imanum/dry024>

4 Other Questions

- Let $f(\lambda) : [0, 1] \rightarrow [0, 1]$ be writable as $f(\lambda) = \sum_{n \geq 0} a_n \lambda^n$, where $a_n \geq 0$ is rational, a_n is nonzero infinitely often, and $f(1)$ is irrational. Then what are simple criteria to determine whether there is $0 < p < 1$ such that $0 \leq a_n \leq p(1-p)^n$ and, if so, to find such p ? Obviously, if (a_n) is nowhere increasing then $1 > p \geq a_0$.
- For each $r > 0$, characterize the functions $f(\lambda)$ that admit a Bernoulli factory where the expected number of coin flips, raised to the power of r , is finite.
- **Multiple-output Bernoulli factories**⁴¹: Let $f(\lambda) : [a, b] \rightarrow (0, 1)$ be continuous, where $0 < a, a < b, b < 1$. Define the entropy bound as $h(f(\lambda))/h(\lambda)$, where $h(x) = -x \ln(x) - (1-x) \ln(1-x)$ is related to the Shannon entropy function. Then there is an algorithm that tosses heads with probability $f(\lambda)$ given a coin that shows heads with probability λ and no other source of randomness (Keane and O'Brien 1994)⁴².

But, is there an algorithm for f that produces multiple independent outputs rather than one and has an expected number of coin flips per output that is arbitrarily close to the entropy bound, uniformly for every λ in f 's domain? Call such an algorithm an *optimal factory*. (See Nacu and Peres (2005, Question 1)⁴³.) And, does the answer change if the algorithm has access to a fair coin in addition to the biased coin?

So far, constants as well as λ and $1 - \lambda$ do admit an optimal factory (see same work), and, as Yuval Peres (Jun. 24, 2021) told me, there is an efficient multiple-output algorithm for $f(\lambda) = \lambda/2$. But are there others? See an **appendix in one of my articles**⁴⁴ for more information on my progress on the problem.

- **Pushdown automata and algebraic functions**⁴⁵: A *pushdown automaton* is a finite state machine with an unbounded stack, driven by a biased coin with an unknown probability of heads, λ . Its stack starts with a single symbol. On each step, the machine flips the coin, then, based on the coin flip, the current state, and the top stack symbol, it moves to a new state (or keeps it unchanged) and replaces the top stack symbol with zero or more symbols. When the stack is empty, the machine stops and returns either 0 or 1 depending on the state it ends up at.

Let $f(\lambda)$ be continuous and map the open interval $(0, 1)$ to itself. Mossel and Peres (2005)⁴⁶ showed that a pushdown automaton can output 1 with probability $f(\lambda)$ only if f is *algebraic over the rational numbers* (there is a nonzero polynomial $P(x, y)$ in two variables and whose coefficients are rational numbers, such that $P(x, f(x)) = 0$ for every x in the domain of f). See an **appendix in one of my articles**⁴⁷ for more information on my progress on the problem.

Prove or disprove:

1. If f is algebraic over rational numbers it can be simulated by a pushdown automaton.
2. $\min(\lambda, 1 - \lambda)$ and $\lambda^{1/p}$, for every prime $p \geq 3$, can be simulated by a pushdown automaton.

⁴⁰<https://www.chebfun.org/ATAP/>

⁴¹<https://mathoverflow.net/questions/412772/from-biased-coins-to-biased-coins-as-efficiently-as-possible>

⁴²Keane, M. S., and O'Brien, G. L., "A Bernoulli factory", *ACM Transactions on Modeling and Computer Simulation* 4(2), 1994.

⁴³Nacu, Șerban, and Yuval Peres. "Fast simulation of new coins from old", *The Annals of Applied Probability* 15, no. 1A (2005): 93-115. <https://doi.org/10.1214/105051604000000549>

⁴⁴https://peteroupc.github.io/bernsupp.html#Multiple_Output_Bernoulli_Factory

⁴⁵<https://cstheory.stackexchange.com/questions/50853/from-coin-flips-to-algebraic-functions-via-pushdown-automata>

⁴⁶Mossel, Elchanan, and Yuval Peres. New coins from old: computing with unknown bias. *Combinatorica*, 25(6), pp.707-724, 2005. With an appendix by Christopher Hillar. <https://doi.org/10.1007/s00493-005-0043-1>

⁴⁷https://peteroupc.github.io/bernsupp.html#Pushdown_Automata_and_Algebraic_Functions

3. Given that f is algebraic over rational numbers, it can be simulated by a pushdown automaton if and only if its “critical exponent” is a dyadic number greater than -1 or has the form $-1 - 1/2^k$ for some integer $k \geq 1$.⁴⁸
- **Coin-flipping degree**⁴⁹: Let $p(\lambda)$ be a polynomial that maps $[0, 1]$ to itself and satisfies $0 < p(\lambda) < 1$ whenever $0 < \lambda < 1$. Then its *coin-flipping degree* (Wästlund 1999)⁵⁰ is the smallest value of n such that p ’s *Bernstein* coefficients of degree n lie in $[0, 1]$. Given that a polynomial’s degree is m and its “standard” coefficients are integers, what are upper bounds (or even exact maximums) on its coin-flipping degree?
- **Simple simulation algorithms**⁵¹: References are sought to papers and books that describe irrational constants in the interval $(0, 1)$ or Bernoulli factory functions (continuous functions mapping $(0, 1)$ to itself) in any of the following ways. Ideally they should involve only rational numbers and should not compute p -adic digit expansions.
 - Simulation experiments that succeed with an irrational probability.
 - Simple **continued fraction**⁵² expansions of irrational constants.
 - Functions written as infinite power series with rational coefficients (see “**Certain Power Series**”⁵³).
 - Irrational numbers written as series expansions with rational coefficients (see “**Certain Converging Series**”⁵⁴).
 - Functions whose integral is an irrational number.
 - Closed shapes inside the unit square whose area is an irrational number. (Includes algorithms that tell whether a box lies inside, outside, or partly inside or outside the shape.) **Example**.⁵⁵
 - Generate a uniform (x, y) point inside a closed shape, then return 1 with probability x . For what shapes is the expected value of x an irrational number? **Example**.⁵⁶
- Given integer $m \geq 0$, rational number $0 < k < \exp(1)$ (and especially $k = 1$ or $k = 2$), and unknown heads probability $0 \leq \lambda \leq 1$, find a **Bernoulli factory**⁵⁷ for—

$$f(\lambda) = \exp(-(\exp(m + \lambda) - (k(m + \lambda)))) = \frac{\exp(-\exp(m + \lambda))}{\exp(-(k(m + \lambda)))}, \quad (\text{PD})$$

that, as much as possible, avoids calculating $h(\lambda) = \exp(m + \lambda) - k(m + \lambda)$, and especially avoids floating-point operations; in this sense, the more implicitly the Bernoulli factory works with irrational or transcendental functions, the better. The right-hand side of (PD) can be implemented by **ExpMinus**⁵⁸

⁴⁸On pushdown automata: Etesami and Yannakakis (“Recursive Markov chains, stochastic grammars, and monotone systems of nonlinear equations”, *Journal of the ACM* 56(1), pp.1-66, 2009) showed that pushdown automata with rational probabilities are equivalent to recursive Markov chains (with rational transition probabilities), and that for every recursive Markov chain, the system of polynomial equations has nonnegative coefficients. But this paper doesn’t deal with the case of recursive Markov chains where the transition probabilities cannot just be rational, but can also be λ and $1 - \lambda$ where λ is an unknown rational or irrational probability of heads. Also, Banderier and Drmota (“Formulae and asymptotics for coefficients of algebraic functions”, *Combinatorics, Probability and Computing* 24(1), pp.1-53., 2014) showed the asymptotic behavior of power series solutions $f(\lambda)$ of a polynomial system, where both the series and the system have nonnegative real coefficients. Notably, functions of the form $\lambda^{1/p}$ where $p \geq 3$ is not a power of 2, are not possible solutions, because their so-called “critical exponent” is not dyadic. But the result seems not to apply to *piecewise* power series such as $\min(\lambda, 1 - \lambda)$, which are likewise algebraic functions.

⁴⁹<https://mathoverflow.net/questions/448538/bounds-on-the-coin-flipping-degree>

⁵⁰Wästlund, J., “**Functions arising by coin flipping**”, 1999.

⁵¹<https://stats.stackexchange.com/questions/541402/what-are-relatively-simple-simulations-that-succeed-with-an-irrational-probability>

⁵²https://peteroupc.github.io/bernoulli.html#Continued_Fractions

⁵³https://peteroupc.github.io/bernoulli.html#Certain_Power_Series

⁵⁴https://peteroupc.github.io/bernoulli.html#Certain_Converging_Series

⁵⁵https://peteroupc.github.io/bernoulli.html#pi_4

⁵⁶https://peteroupc.github.io/bernoulli.html#4_3_pi

⁵⁷<https://peteroupc.github.io/bernoulli.html>

⁵⁸https://peteroupc.github.io/bernoulli.html#ExpMinus_exp_minus_z

and division Bernoulli factories, but is inefficient and heavyweight due to the need to calculate ϵ for the division factory. In addition there is a Bernoulli factory that first calculates $h(\lambda)$ and $\text{floor}(h(\lambda))$ using constructive reals and then runs **ExpMinus**, but this is likewise far from lightweight.

Prove or disprove:

- Given that $f : [0, 1] \rightarrow (0, 1]$ is convex, the polynomials $(g_n) = (B_n(f) - \max_{0 \leq \lambda \leq 1} \text{abs}(B_n(f)(\lambda) - f(\lambda)))$ (where n is a positive integer power of 2) are in Bernstein form of degree n , converge to f from below, and satisfy: $(g_{2n} - g_n)$ is a polynomial with nonnegative Bernstein coefficients once it's rewritten to a polynomial in Bernstein form of degree exactly $2n$. The same is true for the polynomials $(g_n) = (B_n(f) - \text{abs}(B_n(f)(1/2) - f(1/2)))$, if f is also symmetric about $1/2$.
- Let $f : (D \subseteq [0, 1]) \rightarrow [0, 1]$. Given a coin that shows heads with probability λ (which can be 0 or 1), it is possible to toss heads with probability $f(\lambda)$ using the coin and no other sources of randomness (and, thus, f is **strongly simulable**⁵⁹) **if and only if**—
 - f is constant on its domain, or is continuous and polynomially bounded on its domain (*polynomially bounded* means, both f and $1 - f$ are not less than $\min(x^n, (1 - x)^n)$ for some integer n (Keane and O'Brien 1994)⁶⁰), and
 - $f(0)$ is 0 or 1 if 0 is in f 's domain and $f(1)$ is 0 or 1 whenever 1 is in f 's domain, and
 - if $f(0) = 0$ or $f(1) = 0$ or both, then there is a polynomial $g(x) : [0, 1] \rightarrow [0, 1]$ with computable coefficients, such that $g(0) = f(0)$ and $g(1) = f(1)$ whenever 0 or 1, respectively, is in the domain of f , and such that $g(x) > f(x)$ for every x in the domain of f , except at 0 and 1, and
 - if $f(0) = 1$ or $f(1) = 1$ or both, then there is a polynomial $h(x) : [0, 1] \rightarrow [0, 1]$ with computable coefficients, such that $h(0) = f(0)$ and $h(1) = f(1)$ whenever 0 or 1, respectively, is in the domain of f , and such that $h(x) < f(x)$ for every x in the domain of f , except at 0 and 1.

A condition such as “0 is not in the domain of f , or f can be extended to a Lipschitz-continuous function on $[0, \epsilon]$ for some $\epsilon > 0$ ” does not work. A counterexample is $f(x) = (\sin(1/x)/4 + 1/2) \cdot (1 - (1 - x)^n)$ for $n \geq 1$ ($f(0) = 0$), which is strongly simulable at 0 despite not being Lipschitz at 0. ($(1 - x)^n$ is the probability of the biased coin showing zero n times in a row.) Keane and O'Brien already showed strong simulability when D contains neither 0 nor 1.

5 Notes

⁵⁹<https://mathoverflow.net/questions/404961/from-biased-coins-and-nothing-else-to-biased-coins>

⁶⁰Keane, M. S., and O'Brien, G. L., “A Bernoulli factory”, *ACM Transactions on Modeling and Computer Simulation* 4(2), 1994.